INTERNAL TECHNICAL & SYSTEM ANALYSIS DISCUSSION PAPER

10 MW (0.25C) Battery Energy Storage System (BESS)

Audience: System Planning, T&D;, Operations, ProtectionStatus: Internal engineering paper for decision support (source-checked)

1. Executive Summary

This paper consolidates engineering positions for a 10 MW / 40 MWh (0.25C, 4-hour) grid-connected BESS. It formalizes control modes, quantitative performance targets, model requirements, and verification steps. PQ and droop modes are mandatory for grid-connected operation; VF (grid-forming) is optional/beneficial and deployed selectively based on system studies. Targets reflect international practice (IEEE 1547 family) and proven deployments of grid-forming BESS (Hornsdale VMM; ESCRI-SA) and microgrids (Snohomish Arlington).

2. System Context & Objectives

The grid is integrating higher shares of inverter-based resources (IBR). The 10 MW BESS is expected to: (i) provide primary frequency response; (ii) supply reactive support and voltage regulation at the point of connection (POC); (iii) ride through credible faults and disturbances; and (iv) avoid adverse control interactions with existing IBR and synchronous plant. Day-to-day use emphasises grid-following operation with droop-based support. Grid-forming operation (VF) is considered an option for weak/critical nodes or where resilience objectives justify added complexity.

3. Control Architecture & Operating Modes

The PCS implements inner current control with outer power/frequency and voltage/reactive loops, supervised by coordinated limiters and protection. Three canonical operating modalities are defined for this programme:

Mode	Control Objective	Status / Deployment Guidance
VF (grid-forming)	Regulate bus V and f; controlled short-circuit contribution; fast AVR as offered	Optional / Beneficial - apply at weak/critical nodes or when studies show net stability/resilience benefit
PQ (grid-following)	Track P*, Q* at POC under PLL synchronization	Mandatory - default day-to-day operating mode
Droop (P-f & Q-V)	Provide primary frequency and voltage/reactive support; enable load-sharing	Mandatory - required for stability and local support

4. Quantitative Performance Targets (POC/SCR-tuned)

Function	Requirement / Target (internal; aligned to Appendices numerical standards)
P-f droop (primary frequency)	Droop 1-9% (4% default); configurable deadband; response ≤ 0.2 s

Function	Requirement / Target (internal; aligned to Appendices numerical standards)
Q-V droop / VAR control	Stable Q-V droop; V, Q, or PF modes at POC; tune with documented stability margins
Frequency withstand	Continuous operation across utility 50-Hz bands used internally; avoid nuisance tripping
RoCoF tolerance	Retain synchronism for ramps up to 4 Hz/s (250 ms avg) and 2.5 Hz/s (500 ms) in staging tests
LVRT/HVRT ride-through	Ride-through per internal voltage bands; reactive current injection during dips; active P recovery target ≥95% in ~100 ms post-clear
Reactive current injection (faults)	Provide ~2% reactive current for each 1% drop in retained voltage during faults (≈2 pu/pu k-factor) - internal standard
Set-point timing	Start ≤ 50 ms; achieve ≤ 300 ms; settle ≤ 400 ms (non-oscillatory)
Power quality	Harmonics, flicker, unbalance within specified limits at POC
Reactive capability	≥ ±0.3 · Pn at POC (four-quadrant)
Over-current envelope	Declare OEM-validated continuous and short-time current limits; claims above ~120% for ≥60-120 s require thermal/oversizing justification

5. POC Grid Strength & SCR Criteria

Screen the point of connection (POC) short-circuit ratio (SCR). Internal appendices specify stability under low SCR for optional VF operation and prudent PLL tuning for GFL operation:

Operating Mode / Study Case	Minimum Grid Strength Target & Notes
Grid-following (PQ + droop, default)	Design/tune for robust operation at SCR ≥ ~3; validate small-signal margins and PLL bandwidth limits.
Grid-forming (VF, optional)	Demonstrate stable, controllable operation at minimum SCR ≈ 1.0 with current limiting; show seamless VF↔GFL transition (if offered).
Mixed IBR environments	Verify no adverse interactions; if POD is offered, demonstrate damping of 0.2-2.5 Hz modes.

6. Formal Control Definitions

- P-f droop: $\Delta P = -K_{pf} \cdot \Delta f$; Kpf from droop (%) and Pn; include rate limiting & anti-windup.
- Q-V droop: $\Delta Q = +K_{qv} \cdot \Delta V$; define reactive priority; demonstrate small- and large-signal stability margins.
- VF option: Outer V,f regulators with inner current limiting; controlled short-circuit contribution; fast AVR where implemented.
- PLL & synchronization (GFL): robust under low inertia; declare lock range, RoCoF tracking, and recovery dynamics.
- Limiter coordination: coordinate P/Q limiters with priority to avoid wind-up and mode chattering.

7. Protection & Fault Behaviour (POC Coordination)

- Fault current: declare peak/AC components and duration with limiting strategy; coordinate with breaker ratings.

- Trip logic: trip only to protect equipment integrity; otherwise ride-through; document thresholds.
- Communications loss: define fail-safe (hold last / set-point ramp-down) and recovery logic.

8. RMS & EMT Models - Requirements

- Provide RMS (phasor) and EMT models with functionally equivalent behaviour; declare tool versions and solver/time-steps (typical: RMS 5-10 ms; EMT 25-50 µs).
- Include control blocks: P-f droop; Q-V droop/AVR; LVRT/HVRT; reactive current injection; PLL. Optional: virtual inertia, POD, VF↔GFL seamless transition.
- Parameter spreadsheets (units, min/max, tuned values); include current limiters and thermal protection tied to the over-current envelope.
- Mandatory benches: frequency step (±0.2 Hz); voltage step (±5%); LVRT/HVRT sequences; RoCoF ramps; near/far faults; optional POD injection.
- Equivalence evidence: overlay RMS vs EMT for identical cases; provide run scripts/projects and plots.

9. Verification & Acceptance

- Type/FAT: certified reports for droop timing, voltage/reactive support, ride-through and recovery timings, and power quality spectra.
- SAT: witnessed tests for set-point timing, LVRT/HVRT sequences, reactive injection and recovery, RoCoF tolerance; verify SCADA/AGC/AVC interoperability.
- Non-conformance: vendor to correct and resubmit models/firmware with change logs and comparative plots.